

# Scalability & Future Plan

|               |                                                         |
|---------------|---------------------------------------------------------|
| Date          | 29 September 2026                                       |
| Team ID       | SWTID-2026-9766                                         |
| Project Name  | ComicCraft - AI Comic Story Creator using Gemini Models |
| Team Size     | 4                                                       |
| Team Leader   | Priya                                                   |
| Maximum Marks | 1 Mark                                                  |

**Scalability & Future Plan:**

This document outlines how the current ComicCraft solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

**Current System Limitations:**

| S.No | Limitation                                                                                                 | Impact                                                                                  | Priority to Address |
|------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------|
| 1    | Current version is mainly focused on single comic generation and does not include user account management. | Limits personalized storage and long-term access to generated comics.                   | High                |
| 2    | AI-based story and image generation can require significant processing time and resources.                 | May affect response time when usage increases.                                          | High                |
| 3    | The current deployment is designed primarily for local use.                                                | Large-scale public access and distributed deployment require additional infrastructure. | Medium              |

**Scalability Plan:**

| S.No | Scalability Aspect | Current State                                                             | Proposed Upgrade / Solution                                                                                                                              |
|------|--------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Designed mainly for local/small-scale usage.                              | Deploy the FastAPI application on scalable cloud infrastructure and use load balancing or multiple application instances for increased concurrent users. |
| 2    | Data Storage       | Generated comics and PDF files are stored within the project environment. | Introduce cloud/object storage and a database for user profiles, comic metadata, generated content, and downloadable files.                              |
| 3    | Performance        | AI generation is performed during the comic creation workflow.            | Use asynchronous/background processing, caching, optimized model usage, and queue-based processing to reduce waiting time under heavy load.              |
| 4    | Security           | API keys are configured through environment variables.                    | Add authentication, authorization, secure API-key management, input validation, rate limiting, HTTPS, and monitoring for production deployment.          |

**Future Roadmap:**

| Phase   | Planned Feature / Enhancement                                                               | Target Timeline     | Expected Impact                                                       |
|---------|---------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------|
| Phase 1 | User accounts and profiles; save and manage generated comics.                               | Short Term          | Personalized experience and persistent access to user-created comics. |
| Phase 2 | Cloud deployment, database integration, and scalable file storage.                          | Medium Term         | Support for increased users, data, and reliable access.               |
| Phase 3 | Advanced comic customization, multi-page story arcs, and comic libraries.                   | Medium to Long Term | Richer storytelling and improved customization.                       |
| Phase 4 | Performance optimization, monitoring, stronger security, and expanded AI/model integration. | Long Term           | More efficient, secure, and extensible ComicCraft platform.           |